Comments on
Functional Inequalities on the Quantum Boolean Cube with Applications to Quantum Learning Theory

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Analysis on the dyadic group $\{-1,1\}^n$ (i.e. Boolean cube in the present paper) and its Banach-space-valued generalization have important applications in harmonic analysis, probability theory, and functional analysis. This paper is mainly concerned with developing its noncommutative theory. A natural noncommutative counterpart of dyadic group $\{-1,1\}^n$ is the so-called CAR algebra. The analysis on the CAR algebra goes back at least to [1], [CL93] and [LP98] . Another noncommutative counterpart of the dyadic group $\{-1,1\}^n$ is the matrix algebra $\mathbb{M}_{2^n} := \mathbb{M}_2^{\otimes n}$ (i.e. quantum Boolean cube). This paper focuses on the analytic theory on the matrix algebra $\mathbb{M}_{2^n}$ and its applications.

In my preliminary assessment, the main contributions of this paper are as follows:

(1) It is proved that the quantum Ornstein-Uhlenbeck semigroup is holomorphic on $L_p(\mathbb{M}_{2^n})$, $1 < p < \infty$.

In the classical setting, this result was established by Pisier in his remarkable paper [Pi82a]. To state this in detail, let us also recall basic notation. Note that every function $f : \{-1,1\}^n \to X$ admits the Walsh-Fourier expansion

$$f(\varepsilon) = \sum_{A\subseteq[n]} \widehat{f}(A) w_A(\varepsilon), \quad \varepsilon = (\varepsilon_1, \cdots, \varepsilon_n) \in \{-1,1\}^n, \tag{1}$$

where $(X, \|\cdot\|)$ is a Banach space, $[n] = \{1, 2, \cdots n\}$ and $(w_A)_{A\subset[n]}$ Walsh-Fourier system in $L_2(\{-1,1\}^n)$. Define

$$\Delta(w_A) = |A| w_A, \quad A \subset [n].$$

Then the corresponding heat semigroup is defined by

$$e^{-t\Delta} f = \sum_{A\subset[n]} e^{-t|A|} \widehat{f}(A) w_A.$$

Here, $|A|$ denotes the volume of A. Pisier's remarkable result states that the semigroup $(e^{-t\Delta})_t$ is holomorphic on the vector-valued Lebesgue space $L_2(\{-1,1\}^n, X)$ if and only if X is K-convex. Since then, Pisier's theorem has found numerous applications, including in Boolean analysis. For applications related to the subject of the present paper, see [MN14, EI20b, Vol24].

In this paper, the authors established corresponding holomorphic extension theorem for quantum Ornstein-Uhlenbeck semigroups, see Theorem 3.1 in Chapter 3. It is of considerable interest to establish a noncommutative analogue. This point is further illustrated in Chapters 4 and 5 of the present paper. More precisely, this result is used to study the heat smoothing estimates and Bernstein-Markov type inequalities on the quantum boolean

cube $\mathbb{M}_{2^n}$. Although the proof of Theorem 3.1 in Chapter 3 is inspired by Pisier's result, it incorporates several novel ideas, including an equivalent representation of the semigroup. Moreover, to the best of my knowledge, relatively few works have addressed the holomorphicity of noncommutative semigroups. It is also worthwhile to investigate the holomorphicity of this specific noncommutative semigroup.

We next turn to a brief discussion of Chapters 4 and 5. As mentioned above, these two chapters mainly concern the heat smoothing estimates and Bernstein-Markov type inequalities on the quantum boolean cube $\mathbb{M}_{2^n}$. These two chapters are motivated by [3, 4]. To overcome the difficulties arising from noncommutativity, however, the authors employed on recent developments in noncommutative analysis, including representation theory for noncommutative semigroups and Cowling's theorem, see Proposition 4.5 (in Chapter 4) and Theorem 5.8 (in Chapter 5).

(2) Several correlation inequalities are proved in the quantum Boolean cube setting. The authors first introduce the notion of monotonically increasing element in the Boolean cube $\mathbb{M}_{2^n}$ via the quantum Walsh-Fourier system, and then establish the quantum analogues correlation inequalities of Harris–Kleitman,Talagrand [Tal96], and Keller–Mossel–Sen [KMS14]. One of the most interesting feature of these inequalities is that the quantum Talagrand inequality correlation differs substantially from its classical counterpart. Recall that Talagrand [Tal96] proved that for monotone increasing functions $f, g : \{-1, 1\}^n \to \{-1, 1\}$, there exists an absolute constant $C > 0$ such that

$$\mathrm{Cov}(f, g) := \mathbb{E}(fg) - \mathbb{E}(f)\mathbb{E}(g) \geq C\Psi\Big(\sum_{j\in[n]} \|D_j(f)\|_{L_1} \|D_j(g)\|_{L_1} \Big), \tag{2}$$

where $\Psi(t) = t/\log(\frac{e}{t})$. However, as the authors demonstrate in Example 7.6 (in Chapter 7), a straightforward generalization of this result to the quantum setting is not possible. Instead, the authors introduce a new quantity $\mathcal{W}_1(T_1, T_2)$ (see eq. (7.11) of Chapter 7), which provides a lower bound in the desired correlation inequality. We note that the new quantum Talagrand correlation inequality (Theorem 7.10 in Chapter 7) is established under a so-called similarity assumption on $T_1, T_2 \in \mathbb{M}_{2^n}$, inspired by the work in [KKM16].

(3)

My suggestions are listed below

1.

References

[1] S. Ayupov, S. Ferleger, and F. Sukochev. Isomorphism between the associative and non-associative L_p-spaces of type II_1 hyperfinite factors. Mathematica Scandinavica (1996): 271-285.